

SECTION—A

1. What is Microprogram? Differentiate between Hardwired and Micro programmed control. 5
2. Discuss the evolution of Computer architecture. 5
3. Write the steps of fetch and decode and instruction. 5
4. Explain various types of Registers used for designing basic Computer. 5
5. Define the Addressing modes. What are the various types of addressing modes? 5

SECTION—B

6. What are the various modes of Data transfer? 5
7. Define parallel processing. Explain pipelining with an example. 5

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8. What are the types of Mapping organizations of cache memory? 5
9. Explain DMA with the help of block diagram. 5
10. Explain cache memory in detail. Also compare it with auxiliary memory. 5

SECTION—C

11. Answer the following questions : 10×2=20

- (i) Differentiate between RAM and ROM.
- (ii) What do you mean by Instruction format?
- (iii) What is main memory?
- (iv) What is the function of accumulator?
- (v) Write a short note on control memory.
- (vi) What is program counter?

- (vii) Define register transfer language.
- (viii) What is Micro program sequencer?
- (ix) Write a short note on Virtual memory.
- (x) Define the CPI.

G-13/2059
COMPUTER SYSTEM ARCHITECTURE—210
Semester—IV

Time Allowed : 3 Hours]

[Maximum Marks : 50

Note :— The candidates are required to attempt *three* questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of *ten* short answer type questions carrying 2 marks each.

SECTION—A

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|----|--|---|
| 1. | Discuss various components of computer. | 5 |
| 2. | Explain shift microoperations in detail. | 5 |
| 3. | Discuss control memory address sequencing. | 5 |
| 4. | Discuss hardwired programmed control. | 5 |
| 5. | Give the flowchart for addition of binary numbers. | 5 |

SECTION—B

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|-----|--|---|
| 6. | Explain memory hierarchy in detail. | 5 |
| 7. | Explain fully associative and set associative cache. | 5 |
| 8. | Compare memory driven I/O and program driven I/O. | 5 |
| 9. | Discuss various types of pipelining in detail. | 5 |
| 10. | Discuss any two I/O buses. | 5 |

SECTION—C

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|-----|--------------------------------------|---|
| 11. | (i) Explain the role of ALU. | 2 |
| | (ii) Name various types of buses. | 2 |
| | (iii) Define stack pointer. | 2 |
| | (iv) What is the role of a register? | 2 |
| | (v) Define cache memory. | 2 |
| | (vi) Name various pipeline hazards. | 2 |
| | (vii) Define virtual memory. | 2 |
| | (viii) Explain DMA in brief. | 2 |
| | (ix) Define principle of locality. | 2 |
| | (x) Define main memory. | 2 |

COMPUTER SYSTEM ARCHITECTURE**CPE-201****Semester III**

Time: Three Hours

Maximum Marks: 50

Note: Attempt any six question selecting three questions from each Section A and B. Section C is Compulsory.

Section A

- Q1. Design 4 bit arithmetic circuit using full adders.
- Q2. What is instruction cycle for basic computer? Explain various phases of instruction cycle with detailed flowchart.
- Q3. What is micro program sequencer? Design micro program sequencer for control memory.
- Q4. What is Booth multiplication algorithm? Discuss hardware and flowchart for Booth algorithm.
- Q5. Explain various addressing modes.

 $3 \times 5 = 15$ **Section B**

- Q6. What are the performance considerations for memory organisation?
- Q7. What is mapping? Discuss direct mapping in detail.
- Q8. What is DMA? Discuss DMA transfer in computer system with detailed diagram.
- Q9. What are various pipeline conflicts? Explain any two methods for handling branch difficulties.
- Q10. A computer uses RAM chip of 1024X1 Capacity.
How many chips are needed and how should their address lines be connected to provide a memory capacity of 1024 bytes.

 $3 \times 5 = 15$ **Section C**

Q11

- a. What is program counter?
- b. What is selective set micro operation?
- c. What is locality of reference?
- d. What is bootstrap loader?
- e. Write various algorithms for page replacement in cache memory.
- f. What is address sequencer for micro program control unit?
- g. Draw and discuss typical RAM chip of 128X8.
- h. What is resource conflict in pipeline?
- i. What is priority interrupt?
- j. How computer store negative numbers.

 $2 \times 10 = 20$

Roll No.

Total Pages : 4

11061/NJ**D-18/2111****COMPUTER SYSTEM ARCHITECTURE**

Paper-201

Semester-III

Time Allowed : 3 Hours] [Maximum Marks : 50

Note : The candidates are required to attempt **three** questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

SECTION—A

1. Write the steps to fetch and decode an instruction. 5

2. Explain the Microprogram sequencer. 5
3. Discuss the Evolution of Computer architecture. 5
4. A RAM chip has a capacity of 1024 words and each word is of 1 byte. To construct a 16K * 16 main memory from 1K * 8 RAM. Specify the following terms : 5
 - (i) Size of decoder
 - (ii) No. of decoder
 - (iii) No. of RAM chips.
5. Explain various types of registers used for designing basic computer. 5

SECTION—B

6. What are addressing modes? Explain various addressing modes. 5
7. Show the content of registers using both algorithm during the process of multiplication 5×2 . 5

8. A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with a bus. 5

(i) How many selection inputs are there in each multiplexer.

(ii) What size of multiplexer are needed?

(iii) How many multiplexers are there in bus?

9. Discuss DMA controller with a diagram. 5

10. The non-pipelining system takes 40ns to process a task. The same task can be processes in six segment pipeline with a clock cycle time of 10ns. Determine the speedup ratio of pipeline for 100 tasks. 5

SECTION—C

11. Explain the following questions : 10×2=20

(i) What do you mean by register transfer language?

(ii) How many memory chips are needed to construct $2M * 16$ memory system using $512 K * 8$ static memory chips.

(iii) What is Micro program sequencer?

(iv) Define and explain memory hierarchy.

(v) Write logic equation for binary half adder.

(vi) What are the disadvantages of increasing the number of stages in pipeline processing ?

(vii) What is an instruction format in Computer architecture?

(viii) What is a difference between Hardwired and Microprogrammed control?

(xi) Define the MIPS rate.

(x) What do you mean by control memory?